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Studierichting: Informatica
Oefeningenreeks 4A nr. 42.16

$$\int \frac{2x^3 + 5x^2 - 1}{x^2 + x - 2} dx$$

$$x^2 + x - 2 = (x + 2)(x - 1)$$

$$\frac{2x^3 + 5x^2 - 1}{x^2 + x - 2} = 2x + 3 + \frac{x + 5}{x^2 + x - 2}$$

$$\frac{x + 5}{x^2 + x - 2} = \frac{x + 5}{(x + 2)(x - 1)} = \frac{A}{x + 2} + \frac{B}{x - 1}$$

$$A = \left. \frac{x + 5}{x - 1} \right|_{x=-2} = \frac{3}{-3} = -1$$

$$B = \left. \frac{x + 5}{x + 2} \right|_{x=1} = \frac{6}{3} = 2$$

$$\frac{2x^3 + 5x^2 - 1}{x^2 + x - 2} = 2x + 3 - \frac{1}{x + 2} + \frac{2}{x - 1}$$

$$\int \frac{2x^3 + 5x^2 - 1}{x^2 + x - 2} dx = \int (2x + 3) dx - \int \frac{dx}{x + 2} + 2 \int \frac{dx}{x - 1}$$

$$= x^2 + 3x - \ln |x + 2| + 2 \ln |x - 1| + c$$

References